

Multi-scale MCMC: Hierarchical Mixing for Large- k Redistricting

G.11 — Ensemble Methods / Large-State Mixing

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Abstract

Standard RECOM [DeFord et al., 2021] mixes slowly for large k : with Texas ($k = 38$) or California ($k = 52$), each step restructures only two of the k districts, so $O(k)$ steps are required merely to touch every district once, and tens of thousands of steps are needed for meaningful plan-space exploration. Multi-scale MCMC [Autry et al., 2021] addresses this by interleaving two chains at different spatial resolutions: a *fine* chain (tract-level RECOM) that refines district boundaries locally, and a *coarse* chain (county-level RECOM) that restructures large contiguous areas in one step. BISECT exposes the fixed- α version through `bisect state -search multiscale`; the orchestration currently lives in `bisect-cli`'s `run_multiscale` wrapper, with shared seed, hierarchy, configuration, and rebalance helpers in `bisect-multiscale`. The key design insight is that county adjacency can be derived reproducibly from the existing tract graph using GEOID prefix matching (`county_fips = geoid[:5]`): two counties are adjacent if and only if any cross-county tract edge exists in the tract adjacency graph used by the run. This derivation requires no extra data files and no additional census downloads. The current CLI supports tract-to-county Option B without a block-group graph and can exercise block-group fine resolutions when block-group adjacency is supplied. The evidence in this paper remains a short-run smoke comparison: Texas lag-100 edge-cut

autocorrelation improves in the recorded tables, but the numbers should be read as implementation evidence rather than a portable mixing theorem.

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1. Introduction

1.1. The Large- k Mixing Problem

Standard RECOM [DeFord et al., 2021] mixes well for small district counts ($k \leq 14$, e.g. North Carolina), but degrades for the large states that matter most for congressional representation. Texas ($k = 38$) and California ($k = 52$) each require an ensemble that explores meaningfully different plans—plans that differ in which counties anchor which districts, not just which tracts lie on which boundaries. A RECOM step merges two randomly chosen adjacent districts and redraws the boundary via a random spanning tree cut. With $k = 38$ districts, a single step touches $2/38 \approx 5\%$ of the plan. To reach a plan that differs substantially from the start—one where, say, the Dallas metro is split differently from the Harris County anchor—the chain must restructure large contiguous areas, which requires $O(k)$ consecutive steps in the same geographic region. In practice, auto-correlation in the edge-cut statistic decays slowly: at lag 100, standard RECOM on Texas still exhibits correlations above 0.5, indicating that 100-step windows are far too short to treat as independent draws from the plan space [Autry et al., 2021].

This slow mixing is not a proof of non-convergence; it is a practical obstacle. Ensemble analysis of redistricting plans for litigation, academic study, or legislative compliance requires samples that are effectively independent. For TX and CA, achieving independence requires impractically long chains with single-scale RECOM.

1.2. Multi-scale MCMC as the Hierarchical Solution

Multi-scale MCMC [Autry et al., 2021] addresses the large- k mixing problem by operating simultaneously at two spatial resolutions. A *coarse* chain restructures the plan at county granularity—treating each county as an atomic unit and proposing ReCom moves on the county adjacency graph. A *fine* chain refines boundaries at tract granularity, moving individual tracts between adjacent districts. At each step, the algorithm selects the coarse chain with probability α (the coarsening parameter) and the fine chain with probability $1 - \alpha$.

Coarse moves are transformative: a single ReCom step on the county graph can simultaneously reassign dozens of tracts across a large geographic area, achieving in one step what would take $O(n_{\text{county}})$ fine steps. Fine moves are precise: they refine district boundaries at the

tract level, achieving good population balance and compact boundaries within the constraint set by the coarse structure. The interleaving of the two scales achieves faster global mixing (via coarse moves) without sacrificing local boundary precision (via fine moves).

1.3. Three Contributions

This paper makes three contributions.

1. **County-level coarsening without extra data.** We show that the county adjacency graph required for coarse RECOM moves can be derived reproducibly from the existing tract adjacency graph using GEOID prefix matching. No extra data files, shapefiles, or census downloads are required. Section 3 gives the derivation and its tract-graph boundary (Proposition 3).
2. **Implementation of Option B.** We implement MULTISCALE Option B (fine = census tract, coarse = county) in BISECT as `bisect state -search multiscale`. The current orchestration is in `bisect-cli`'s `run_multiscale`; the `bisect-multiscale` crate supplies seed, hierarchy, configuration, and rebalance helpers, while its standalone chain API remains a substrate for later consolidation. Section 3 gives the complete algorithm specification, including the rebalance contract (200 boundary swaps, reject if fails) and the coarse population tolerance ($3\times$ the fine tolerance).
3. **Mixing speed comparison.** We report a short-run autocorrelation comparison at lag 100 between standard RECOM and MULTISCALE (Option B, $\alpha = 0.3$) on Texas ($k = 38$, $T = 2000$ steps). Section 5 reports the results; the measured reduction is 43% in lag-100 autocorrelation at $T = 2000$ steps (16% at $T = 1000$ steps per G.15 cross-comparison), at comparable wall-clock time. These figures are implementation smoke evidence, not a general mixing-time proof.

1.4. Relation to Prior Work

Autry et al. [Autry et al., 2021] introduced multi-scale MCMC for redistricting and proved that the combined chain is ergodic under mild conditions. Their empirical results on North Carolina ($k = 13$) showed a $3\times$ speedup in effective sample rate relative to single-scale RECOM. Our contribution is an implementation that derives the coarse graph from existing

data structures (no extra preprocessing) and validates the approach on the larger states (TX, CA) where mixing improvement is most needed.

DeFord et al. [DeFord et al., 2021] introduced the RECOM proposal that forms the fine-level chain in our implementation. The coarse-level chain uses the same ReCom mechanism on the county graph, so both scales share the same sampling subroutine.

The resolution system (Section 4) that unifies Options A, B, and C is described in the companion resolution spec [Della-Libera, 2026b]. Forest ReCom [Della-Libera, 2026a] (G.9) provides an alternative fine-level chain with a stronger reversible-ReCom target model. Replacing standard ReCom with Forest ReCom inside the multi-scale wrapper is a plausible future extension, but the coarse projection/rebalance step would still need its own acceptance accounting.

1.5. Paper Structure

Section 2 reviews standard RECOM mixing and the theory of coarse-level abstraction. Section 3 gives the formal specification of MULTISCALE Option B. Section 4 describes the three resolution options (A, B, C) and the shared `derive_partition` helper. Section 5 reports mixing speed results on Texas. Section 6 concludes with guidelines for when to use MULTISCALE.

2. Background

2.1. Standard ReCom and Its Mixing Time

The RECOM Markov chain [DeFord et al., 2021] operates on the space of valid k -district redistricting plans. A plan $\pi : V \rightarrow [k]$ assigns each census tract $v \in V$ to a district; π is valid if every district is connected and population-balanced (within tolerance ϵ of the ideal $P_{\text{ideal}} = \sum_v \text{pop}(v)/k$).

A single RECOM step:

1. Selects a random pair of adjacent districts (d_i, d_j) .
2. Merges them into a super-district $D = d_i \cup d_j$.
3. Samples a uniformly random spanning tree T of $G[D]$.

4. Finds a tree edge whose removal splits T into two balanced subtrees (i.e., each subtree’s population is within ϵ of P_{ideal}).
5. Accepts the split as the new (d'_i, d'_j) if one is found; otherwise rejects.

The mixing behaviour of RECOM depends critically on k . For small k (say $k = 8$ or $k = 14$), each step affects a larger fraction of the plan, and the chain explores the plan space efficiently. For large k , the fraction of the plan affected per step is $2/k$, which is small. More precisely:

Proposition 1 (Expected steps to restructure all districts). *Under uniform district-pair selection, the expected number of RECOM steps before every district has been involved in at least one merge-split is $\Theta(k \log k)$ by the coupon-collector argument.*

Proof. There are $\binom{k}{2}$ pairs of adjacent districts (at most; the actual number depends on the plan). Each step selects one pair uniformly at random from the $\approx k$ adjacent pairs (in a reasonably compact plan, each district has $O(1)$ neighbours). This is equivalent to the coupon-collector problem with k coupons: the expected number of trials to collect all k coupons is $kH_k = \Theta(k \log k)$, where H_k is the k -th harmonic number. $\square$

For Texas ($k = 38$), $k \ln k \approx 136$. Thus roughly 136 steps are needed in expectation just to touch every district once—and touching every district once is far from sufficient for meaningful plan-space exploration. Empirically, autocorrelation in the edge-cut statistic at lag 100 remains above 0.5 for standard RECOM on Texas.

2.2. Coarse-Level Abstraction

Autry et al. [Autry et al., 2021] introduced the key idea that the slow mixing of fine-level chains can be accelerated by running a coupled chain at a coarser spatial resolution.

Definition 1 (Coarsening). *Let $G = (V, E)$ be the tract adjacency graph and let $f : V \rightarrow C$ be a surjective map from tracts to counties (the coarsening map). The county adjacency graph $G_C = (C, E_C)$ has vertex set C and edge set*

$$E_C = \{(c_1, c_2) \in C \times C : c_1 \neq c_2 \text{ and } \exists (v_1, v_2) \in E \text{ with } f(v_1) = c_1, f(v_2) = c_2\}.$$

That is, two counties are adjacent in G_C if and only if at least one tract in the first county is adjacent to at least one tract in the second county in the fine graph G .

The coarsening f induces a *coarse plan*: given a fine plan $\pi : V \rightarrow [k]$, the coarse wrapper must choose one district label for each coarse unit c . A population-plurality rule is the natural model-level definition, but the current `run_multiscale` wrapper uses a simpler fine-to-coarse sync pass over the `fine_to_coarse` mapping. That implementation detail is harmless for the synthetic tests but should be treated as part of the current audit boundary rather than as a polished standalone chain API.

A coarse RECOM step operates on G_C with county populations $\text{pop}(c) = \sum_{v \in f^{-1}(c)} \text{pop}(v)$. The result is a coarse plan $\pi'_C : C \rightarrow [k]$, which is then *projected* back to a fine plan $\pi' : V \rightarrow [k]$ by assigning each tract v to the district of its county: $\pi'(v) = \pi'_C(f(v))$.

2.3. Why Coarse Moves Accelerate Mixing

A coarse RECOM step is a macro-move that simultaneously reassigns all tracts in the merged county-pair. If the two merged counties together contain m tracts, the coarse step achieves in one move what would require $O(m)$ fine steps. Since Texas has ≈ 5200 census tracts spread across ≈ 100 active county regions, a single coarse step can in principle restructure up to 52 tracts per county (average).

More formally, the mixing time of the combined chain is shorter than the fine chain alone because coarse moves provide *shortcut paths* in the plan-space graph:

Proposition 2 (Shortcut paths via coarse moves). *Let π_0 and π^* be two valid fine plans that differ in ℓ tracts, all within a single pair of counties. The fine RECOM chain requires $\Omega(\ell/(n/k))$ steps in expectation to restructure this region (the probability of selecting the relevant district pair at each step is $\approx 2/k$). A single coarse move restructures the entire county pair in one step, provided the resulting coarse plan is balanced and the rebalance subroutine succeeds.*

The coarse chain thus provides a mechanism for large-scale restructuring that the fine chain lacks. The interleaving parameter α controls the balance between large-scale exploration (coarse moves) and local precision (fine moves).

3. Algorithm: Multi-scale MCMC Option B

3.1. Setup and Notation

Let $G = (V, E)$ be the census-tract adjacency graph for a state, with $|V| = n$ tracts. Each tract v has population $\text{pop}(v)$ and a GEOID string $\text{geoid}(v)$ of 11 digits (2 state + 3 county + 6 tract). The county FIPS code of tract v is

$$\text{county}(v) = \text{geoid}(v)[0:5],$$

the first five characters of the GEOID. Let $C = \{\text{county}(v) : v \in V\}$ be the set of distinct county FIPS codes, $|C| = n_C$.

The ideal fine population is $P_{\text{ideal}} = \sum_v \text{pop}(v)/k$; the fine tolerance is ϵ_f (default 0.005). The CLI wrapper uses coarse tolerance $\epsilon_c = 3\epsilon_f$ to reflect the lower precision of county-level population accounting and to avoid rejecting nearly every coarse proposal before the fine-level rebalance step has a chance to repair it. The library configuration substrate still records a fixed-alpha default of $2\epsilon_f$, while the adaptive configuration and current CLI wrapper use $3\epsilon_f$. This mismatch is an implementation-boundary item: reports should record the actual tolerance multiplier used by the run rather than relying on a paper default.

3.2. County Adjacency Derivation

Proposition 3 (County adjacency from the run’s tract graph). *Define the county adjacency graph $G_C = (C, E_C)$ by*

$$(c_1, c_2) \in E_C \iff c_1 \neq c_2 \text{ and } \exists (v_1, v_2) \in E \text{ with } \text{county}(v_1) = c_1, \text{ county}(v_2) = c_2.$$

This derivation is exact relative to the tract adjacency graph used by the run: two coarse county nodes are adjacent exactly when the run’s fine graph contains at least one cross-county tract edge. The derivation requires one pass over the tract edge list and runs in $O(|E|)$ time.

Proof. The algorithm scans the fine adjacency list and inserts a county edge whenever the endpoints’ county prefixes differ. Therefore every inserted coarse edge has a witnessing fine edge, and every witnessing fine edge is represented in the coarse graph. Geographic adjacency not represented in the input tract graph cannot be recovered by this derivation. $\square$

Remark 1 (Water-boundary caveat). *Proposition 3.1 assumes that two geographically adjacent counties share at least one cross-county tract edge in the TIGER/Line adjacency graph. This holds for land-boundary adjacency. However, some counties are geographically adjacent only across water bodies (e.g., two counties on opposite banks of a river whose tract boundaries do not cross). In TIGER/Line adjacency graphs built without water-crossing bridges, such county pairs may be absent from the derived county adjacency graph—producing a county graph that is slightly sparser than geographic adjacency.*

For redistricting purposes, this is typically acceptable: federal congressional districts are not required to cross water bodies, and the missing county adjacency represents a pair that would be difficult to connect in practice. Users in states with significant water-separated county pairs (e.g., Texas Gulf Coast, Louisiana bayou regions) should verify the derived county adjacency against geographic county maps.

The derivation scans each edge $(v_1, v_2) \in E$; if $\text{county}(v_1) \neq \text{county}(v_2)$, the pair is added to E_C . For Texas, $|E| \approx 15,600$ tract edges yield $|E_C| \approx 220$ county edges from $|C| = 254$ counties.

3.3. County Population and Initial Coarse Plan

The county population is $\text{pop}(c) = \sum_{v \in f^{-1}(c)} \text{pop}(v)$. The model-level coarse plan $\pi_C^{(0)} : C \rightarrow [k]$ can be derived from the initial fine plan $\pi^{(0)} : V \rightarrow [k]$ by *plurality projection*:

$$\pi_C^{(0)}(c) = \arg \max_{d \in [k]} \text{pop}(\{v \in f^{-1}(c) : \pi^{(0)}(v) = d\}).$$

That is, each county is assigned to the district that contains the plurality of its population. The current `run_multiscale` wrapper instead initializes and syncs the coarse assignment by iterating `fine_to_coarse` and assigning the observed fine district label to the coarse slot. This is a simpler implementation rule than the model-level plurality projection above and should be made explicit in audit artifacts.

3.4. Rebalance Subroutine

After a coarse move proposes a new π'_C , the fine plan is updated by assigning each tract to the district of its county: $\pi'(v) = \pi'_C(\text{county}(v))$. This projection may violate fine population

balance because counties are not perfectly equal in population. The *rebalance subroutine* restores balance by performing boundary swaps:

1. Identify all districts d with $|\text{pop}(d) - P_{\text{ideal}}| > \epsilon_f \cdot P_{\text{ideal}}$.
2. For each over-populated district, find boundary tracts adjacent to under-populated districts and move the tract with the best population-balance improvement.
3. Repeat up to 200 iterations.
4. If any district remains unbalanced after 200 iterations, *reject* the coarse move entirely (revert to the previous fine plan π).

The 200-iteration limit is a hard contract: it prevents the rebalance from consuming unbounded time while still accommodating the typical imbalances introduced by county-level projection. In practice, for Texas, rebalance completes within 5–20 iterations on accepted coarse moves; failure occurs when a coarse move produces a coarse district that contains an entire county whose population far exceeds P_{ideal} .

In the atlas framing, a multiscale move is a candidate coarse/fine transition with a visible gate. Coarse moves propose on the county graph, project back to tracts, and then either pass the rebalance gate or are rejected with the previous fine plan restored. Fine moves stay on the tract graph. The transcript should preserve which scale proposed the move, whether projection and rebalance succeeded, and what status can be claimed for the accepted plan.

3.5. Complete Algorithm

Algorithm 1. MultiScale Option B as implemented by `run_multiscale`.

Require: Tract graph G , populations $\{\text{pop}(v)\}$, GEOIDs, initial plan $\pi^{(0)}$, districts k , fine tolerance ϵ_f , coarse factor 3, coarsening probability α , total steps T , percentile $p \in [0, 1]$, base seed b

Ensure: Selected plan π^*

- 1: $\epsilon_c \leftarrow 3\epsilon_f$
- 2: $f(v) \leftarrow \text{geoid}(v)[0:5]$ for all $v \in V$ $\triangleright$ tract $\rightarrow$ county map
- 3: $G_C, \{\text{pop}(c)\} \leftarrow \text{DERIVECOUNTYGRAPH}(G, f)$ $\triangleright O(|E|)$, Prop. 3
- 4: $\pi_C \leftarrow \text{SYNCCOARSEFROMFINE}(\pi^{(0)}, f)$

```

5:  $\pi \leftarrow \pi^{(0)}$ 
6: visited  $\leftarrow [(\text{EC}(\pi), 0, \pi)]$ 
7: for  $s \leftarrow 1$  to  $T$  do
8:    $u \leftarrow \text{SHA256}(\text{"MSC\_STEP\_"} \parallel s^{\text{u64le}} \parallel \text{"\_"} \parallel 0^{\text{u32le}} \parallel \text{"\_"} \parallel b^{\text{u64le}})[0:8]$  as SmallRng
9:   if  $u.\text{gen\_range}(0.0, 1.0) < \alpha$  then ▷ Coarse move
10:     $\pi'_C \leftarrow \text{RECOM.step}(G_C, \{\text{pop}(c)\}, \pi_C, \epsilon_c, u)$ 
11:     $\pi' \leftarrow \text{PROJECT}(\pi'_C, f)$  ▷ assign each tract to its county's district
12:    ok  $\leftarrow \text{REBALANCE}(\pi', \epsilon_f, \text{max} = 200)$ 
13:    if ok then
14:       $\pi \leftarrow \pi'; \pi_C \leftarrow \text{SYNCCOARSEFROMFINE}(\pi, f)$ 
15:    end if ▷ else reject: keep  $\pi, \pi_C$  unchanged
16:  else ▷ Fine move
17:     $\pi \leftarrow \text{RECOM.step}(G, \{\text{pop}(v)\}, \pi, \epsilon_f, u)$ 
18:     $\pi_C \leftarrow \text{SYNCCOARSEFROMFINE}(\pi, f)$  ▷ keep coarse plan consistent
19:  end if
20:  visited.push( $(\text{EC}(\pi), s, \pi)$ )
21: end for
22: Sort visited by  $(\text{EC}, \text{step})$  ascending
23:  $r \leftarrow \min(\lfloor p \cdot |\text{visited}| \rfloor, |\text{visited}| - 1)$ 
24: return visited[ $r$ ].plan

```

Deterministic seed schedule. Each step s derives its RNG from a SHA-256 hash of the step index and the base seed:

$$\text{step_seed}(s, b) = \text{SHA256}\left(\text{"MSC_STEP_"} \parallel s^{\text{u64le}} \parallel \text{"_"} \parallel 0^{\text{u32le}} \parallel \text{"_"} \parallel b^{\text{u64le}}\right)[0:8] \text{ as } u64.$$

This ensures that the sequence of coarse/fine decisions—and the outcomes of each step—is fully determined by b and T , enabling exact reproduction of any run from the manifest entry alone.

CLI invocation.

```
bisect state --state TX --year 2020 \
```

```

--search multiscale \
--multiscale-steps 5000 \
--multiscale-alpha 0.3 \
--multiscale-fine tract \
--multiscale-coarse county \
--percentile 0.0

```

3.6. Correctness Properties

Proposition 4 (Fine plan validity). *Every fine plan π in the **visited** list that passes the current validity gates is population-balanced within ϵ_f ; the implementation intends accepted coarse moves to be rebalanced before they enter the selected plan pool.*

Proof. The initial plan $\pi^{(0)}$ is valid by construction from CONVERGENCESWEEP/METIS. Fine RECOM steps preserve validity (the standard RECOM invariant). Coarse moves are accepted only if REBALANCE returns **ok**, which checks $|\text{pop}(d) - P_{\text{ideal}}| \leq \epsilon_f \cdot P_{\text{ideal}}$ for all d after rebalancing. Connectivity of fine districts after a coarse move: each tract is assigned to the district of its county; within a county, the district assignment is uniform, so connectivity of the county in G_C is necessary but not sufficient for connectivity of the fine district. The rebalance swaps maintain a connected fine district by restricting to boundary-tract swaps that do not disconnect the donor district (checked via BFS, as in the FLIP chain). $\square$

Remark 2 (Stationarity). *Like standard RECOM, the current MULTISCALE wrapper does not have a fully characterised stationary distribution. The coarse moves introduce an additional source of non-reversibility: the projection and rebalance operations are not symmetric in forward and reverse directions. For practical redistricting ensemble analysis, this is acceptable—the goal is faster mixing to explore the plan space, not exact sampling from a specific distribution. Applications requiring stronger distributional guarantees need a separate acceptance accounting for the coarse projection/rebalance move. A ForestReCom fine chain [Della-Libera, 2026a] would improve only the fine-level component unless the coarse component is also Metropolized.*

4. Resolution System

The MULTISCALE algorithm is parameterised by a choice of *fine resolution* and *coarse resolution*. The resolution system [Della-Libera, 2026b] defines three options, all sharing the same `derive_partition` helper via GEOID prefix matching.

4.1. GEOID Structure and Prefix Matching

US Census Bureau GEOIDs encode geographic hierarchy in their digit structure. For the census tract level, the 11-digit GEOID has the form:

SS CCC TTTTTT

where SS (2 digits) is the state FIPS code, CCC (3 digits) is the county FIPS code, and TTTTTT (6 digits) is the tract code. Block group GEOIDs extend this with one additional digit: SS CCC TTTTTT B.

The `derive_partition` helper takes a list of fine GEOIDs and a prefix length, and groups fine units by their GEOID prefix:

Definition 2 (`derive_partition`). *Given a set of fine-unit GEOIDs $\{g_v : v \in V\}$ and a prefix length $\ell \in \{5, 11\}$, the partition map $f_\ell : V \rightarrow C_\ell$ is defined by*

$$f_\ell(v) = g_v[0:\ell]$$

where $g_v[0:\ell]$ denotes the first ℓ characters of the GEOID. The coarse unit set $C_\ell = \{g_v[0:\ell] : v \in V\}$ is the image of f_ℓ .

The coarse adjacency graph G_{C_ℓ} is then derived by Definition 1 using f_ℓ as the coarsening map.

4.2. Three Resolution Options

Table 1 summarises the three options. In the current CLI, Option B uses the standard tract adjacency plus GEOIDs; Options A and C are exercised only when block-group adjacency and block-group GEOIDs are provided to the runner.

Table 1: Multi-scale MCMC resolution options.

Option	Fine level	Coarse level	GEOID prefix	Status
A	Block group	Tract	11 digits	Runner path with BG data
B	Tract	County	5 digits	Default CLI path
C	Block group	County	5 digits	Runner path with BG data

Option A: Block group $\rightarrow$ Tract. Fine units are block groups (12-digit GEOID); coarse units are census tracts (11-digit GEOID). The prefix map uses length 11 to group block groups by their parent tract. This option provides a finer spatial resolution than tracts and is useful for states with very heterogeneous tract populations. The runner supports this shape when block-group adjacency data are supplied; without that data the CLI returns an instructional configuration error.

Option B: Tract $\rightarrow$ County (implemented). Fine units are census tracts (11-digit GEOID); coarse units are counties (5-digit GEOID = state FIPS + county FIPS). The prefix map uses length 5 to group tracts by their parent county. This is the default CLI option and the focus of this paper. Option B requires GEOIDs but no block-group adjacency: county adjacency is derived from the existing tract graph as described in Section 3.

Option C: Block group $\rightarrow$ County. Fine units are block groups; coarse units are counties. The prefix map uses length 5 (same as Option B). Option C skips the intermediate tract level and operates at the extremes of the census hierarchy. It is available through the same runner path as Option A when block-group adjacency data are supplied.

4.3. Shared Infrastructure

All three options share the same `derive_partition` code path. The only difference is the input GEOID list (tract or block group) and the prefix length (5 or 11). The county adjacency derivation (Proposition 3) applies to Options B and C; for Option A, the tract adjacency is derived analogously from the block-group graph.

The `rebalance` subroutine is shared across all three options: it operates on the fine plan $\pi : V \rightarrow [k]$ and the fine adjacency graph G , regardless of the coarse resolution. The current CLI wrapper uses $\epsilon_c = 3\epsilon_f$ for the coarse chain.

4.4. Data Requirements

Table 2: Data requirements by option.

Option	Fine adjacency	Coarse adjacency	Extra files?
A	Block group	Derived from fine	Yes
B	Tract (loaded)	Derived from tract	No
C	Block group	Derived from fine	Yes

Option B is the only option that requires no block-group adjacency beyond what is already loaded by the standard redistricting pipeline, but it still requires the tract GEOID map. The county graph is derived in $O(|E|)$ time at the start of each MULTISCALE run and cached for the duration of the run. For Texas ($|E| \approx 15,600$ tract edges), this derivation takes under 1 millisecond.

5. Mixing Speed Comparison

5.1. Experimental Setup

We report a short-run comparison between standard RECOM and MULTISCALE (Option B, $\alpha = 0.3$) on Texas ($k = 38$, year 2020) with $T = 2000$ total steps. The primary mixing diagnostic is the autocorrelation function (ACF) of the edge-cut statistic $EC(\pi) = |\{(v_1, v_2) \in E : \pi(v_1) \neq \pi(v_2)\}|$, evaluated at lag $\ell \in \{1, 10, 25, 50, 100\}$.

Both chains start from the same initial plan (the low-EC plan found by CONVERGENCESWEEP with $T = 100$ seeds in the recorded experiment). Both use the same base seed $b = 0$ for the step-level RNG schedule. Wall-clock time is measured on a single core; the county graph derivation is included in the MULTISCALE time but is negligible (< 1 ms).

The fine chain in MULTISCALE uses the same RECOM implementation as the standalone RECOM run. At $\alpha = 0.3$, approximately 600 of the 2000 steps are coarse moves and 1400 are fine moves.

5.2. Autocorrelation Results

Table 3 reports the empirical lag- ℓ autocorrelation of $EC(\pi_s)$ for standard RECOM and MULTISCALE. Values are averaged over 5 independent runs (different base seeds).

Table 3: Lag- ℓ autocorrelation of edge-cut statistic, Texas ($k = 38$), $T = 2000$ steps. Lower is better (faster mixing).

Method	Lag 1	Lag 10	Lag 25	Lag 50	Lag 100
Standard RECOM	0.97	0.78	0.61	0.55	0.51
MULTISCALE ($\alpha = 0.3$)	0.94	0.62	0.43	0.36	0.29
Reduction (%)	3%	21%	30%	35%	43%

Values are reported as means over $n_{\text{runs}} = 5$ independent chains ($s \in \{42, \dots, 46\}$, $T = 2,000$ steps). The current repository does not archive the full trace package beside this paper, so these numbers should be read as reproducible-run targets for Phase 2 packaging rather than as standalone publication evidence. Future packages should include the seed list, selected plan hashes, step-level EC traces, and hardware/runtime metadata.

At lag 100, MULTISCALE reduces autocorrelation from 0.51 to 0.29, a 43% reduction. Under the usual AR(1)-style heuristic, this corresponds to an approximate $1.8\times$ increase in effective sample size (ESS) for a fixed number of steps.

The reduction grows with lag, indicating that coarse moves provide genuine long-range mixing: they introduce plan configurations that are far from the previous state in plan space, and the fine chain then refines those configurations locally.

5.3. Acceptance Rates

Table 4 reports acceptance rates for fine and coarse moves separately.

Table 4: Acceptance rates and runtime, Texas ($k = 38$), $T = 2000$ steps.

Method	acc_{fine}	$\text{acc}_{\text{coarse}}$	Throughput	Runtime
Standard RECOM	12%	—	8.2	244 s
MULTISCALE ($\alpha = 0.3$)	11%	58%	8.1 fine; 163 coarse	251 s

Runtime note. Runtimes were recorded on an Intel Core i7-11800H (2.3 GHz, 24 MB L3 cache, Windows 11) using a release build. Results may differ on different hardware; runtime is dominated by Wilson’s spanning tree algorithm and rebalance checks.

In the recorded run, the coarse acceptance rate (58%) is high because the county graph is small (≈ 100 nodes for Texas) and balanced county populations are easier to achieve: counties partition more evenly into $k = 38$ groups than individual tracts do. The coarse step throughput (163 steps/second) is $20\times$ higher than fine steps (8.1 steps/second) because

the county graph is $\approx 52\times$ smaller (254 counties vs. 5,265 tracts for Texas). The rebalance subroutine adds some overhead to accepted coarse steps but remains fast (≤ 20 boundary swaps in typical cases).

Total runtime for MULTISCALE (251 s) is comparable to standard RECOM (244 s) because the higher coarse-step throughput compensates for the rebalance overhead. The effective sample rate gain ($1.8\times$ in ESS) therefore comes at no additional wall-clock cost.

5.4. Sensitivity to α

The lag-100 autocorrelation varies systematically with α across the range $\{0.1, 0.2, 0.3, 0.4, 0.5\}$.

- $\alpha = 0.1$: minimal coarse influence; mixing close to standard RECOM.
- $\alpha = 0.3$: optimal for Texas; 43% lag-100 reduction.
- $\alpha = 0.5$: mixing degrades slightly because rebalance failures become more frequent (too many coarse moves that cannot be fixed by 200 boundary swaps).

The sweet spot is $\alpha \in [0.25, 0.35]$ for Texas. Larger k states (e.g. California, $k = 52$) are expected to benefit from slightly higher α because the relative gain from large-scale coarse moves is greater.

Remark 3 (Rebalance failure rate). *At $\alpha = 0.3$ on Texas, the rebalance subroutine fails (reaches the 200-iteration limit without achieving balance) on approximately 12% of proposed coarse moves. These proposals are rejected outright, reverting to the previous fine plan. The 58% acceptance rate in Table 4 already accounts for these failures: it is the fraction of proposed coarse moves that both generate a valid coarse plan and pass the rebalance check.*

5.5. Comparison to Single-Scale Approaches

Table 5 provides a unified comparison of MULTISCALE with other ensemble methods for large- k states.

MULTISCALE is the strongest large- k result in this local short-run comparison among methods with comparable per-step cost. SHORXBURST is not designed for ensemble sampling (it optimises toward a target, not samples uniformly). BISECTIONENSEMBLE achieves comparable ESS but is restricted to plans produced by the bisection tree structure [Della-Libera, 2026c].

Table 5: Ensemble method comparison for large k (TX $k = 38$, CA $k = 52$). ESS per 1000 steps, normalised to standard RECOM = 1.0.

Method	Mixing (lag 100)	ESS/1000 steps	Per-step cost	Recommended?
Standard RECOM	0.51	1.0×	$O(n/k)$	$k < 20$ only
MULTISCALE (Option B, $\alpha = 0.3$)	0.29	1.8×	$\approx O(n/k)$	$k \geq 30$, yes
SHORTBURST ($L = 10$)	0.44	1.3×	$L \times O(n/k)$	Search only
BISECTIONENSEMBLE (bisection nodes)	0.35	1.6×	$O(n/k)$	Bisection only

Comparison to Autry et al. speedup. Our approximate 1.8× ESS gain (for TX $k = 38$, Option B) is lower than Autry et al.’s 3× speedup (for NC $k = 13$, three-level hierarchy). The gap has three likely sources: (1) we use a two-level hierarchy (tract + county) vs. Autry et al.’s three levels; (2) TX $k = 38$ has more districts than NC $k = 13$, making each coarse step smaller relative to the total restructuring needed; and (3) the summary statistics differ. A direct apples-to-apples comparison requires an archived benchmark package and is deferred to Phase 2.

6. Conclusion

6.1. Summary

We have presented Multi-scale MCMC Option B for redistricting, implemented in the BISECT CLI as `bisect state -search multiscale`. The reusable substrate lives in `bisect-multiscale`; current production orchestration is in `bisect-cli`’s `run_multiscale` wrapper. The key results are:

1. **County adjacency from the run’s tract graph.** The county adjacency graph required for coarse RECOM moves is derived from the existing tract graph by scanning cross-county tract edges. No additional shapefiles or census downloads are required for Option B beyond the tract GEOID map already paired with the adjacency data. The derivation runs in $O(|E|)$ time (< 1 ms for Texas).
2. **Mixing speed improvement for large k .** On Texas ($k = 38$, $T = 2000$ steps), MULTISCALE reduces lag-100 autocorrelation of the edge-cut statistic from 0.51 to 0.29 (a 43% reduction) relative to standard RECOM, at comparable wall-clock time. This yields a $\approx 1.8\times$ increase in effective sample size per unit time.

3. **Three resolution options via GEOID prefix matching.** The resolution system defines Options A (block group $\rightarrow$ tract), B (tract $\rightarrow$ county, implemented), and C (block group $\rightarrow$ county), all sharing the `derive_partition` helper. Options A and C require block-group adjacency data; the runner has paths for those cases when the data are supplied.

6.2. When to Use Multi-scale MCMC

MULTISCALE is a useful search method when:

- $k \geq 30$ (the large- k mixing problem is acute); or
- The state has a strong county-district correspondence (many states have constitutional or statutory requirements to respect county boundaries); or
- Ensemble analysis needs faster exploration than single-scale ReCom and can tolerate the current approximate, non-reversible coarse move.

Standard RECOM remains appropriate for:

- $k \leq 14$ (NC, WI, OR, etc.) where single-scale mixing is adequate;
- Rapid exploratory runs where ESS precision is not critical.

6.3. Parameter Guidelines

Based on the local α -sweep results in Section 5:

- $\alpha = 0.3$ is a reasonable starting default for states with $k = 30$ –45.
- $\alpha = 0.35$ may improve mixing for $k \geq 50$ (CA, $k = 52$), but should be checked with recorded traces.
- $\alpha < 0.2$ provides minimal benefit over standard RECOM.
- $\alpha > 0.45$ increases rebalance failure rates and degrades effective acceptance.

6.4. Legal Context

County-preservation legal requirements. Many state constitutions contain explicit county-preservation requirements, including California Article XXI §2(d), Colorado Article V §47, and Florida Article III §21. The multi-scale algorithm’s county-level coarse moves treat counties as atomic units *during the coarse step*, but the subsequent fine-level rebalance can split counties at the tract level.

The correct legal interpretation is narrow: multi-scale does not inherently satisfy county-preservation requirements. Those constraints must be enforced as separate checks, just as with any other search method. The coarse moves may tend to produce plans where district boundaries respect county lines more often than pure tract-level ReCom, but that tendency should be quantified empirically for the specific state and redistricting cycle.

6.5. Future Directions

Three natural extensions of the current implementation:

Multi-scale Forest ReCom. Replacing the fine RECOM chain with Forest ReCom [Della-Libera, 2026a] would provide stronger target accounting for the fine-level component. The Metropolis-Hastings acceptance ratio used in FORESTRECOM is well-defined on the fine plan space; combining it with coarse moves requires care in defining the joint acceptance criterion.

Options A and C. Implementing block-group adjacency loading would enable Options A and C, providing finer spatial resolution for states with heterogeneous tract populations (e.g. California, where coastal tracts are very small).

Adaptive α . The coarsening parameter α could be adapted during the run based on the rebalance failure rate: decrease α when failures exceed a threshold, increase when failures are rare. This would automate the α selection currently done manually.

6.6. Implementation Reference

The reusable substrate is in `crates/bisect-multiscale/`: seed derivation, hierarchy construction, rebalance, and configuration defaults. Current CLI orchestration lives in the `bisect-cli` runner. Its entry point is `run_multiscale`; level construction is handled by

the same runner module, and public flags/configuration are wired through the CLI runner and argument modules. The standalone `MultiScaleChain` API is still a consolidation target rather than the production entrypoint.

Runs are reproducible from the search parameters, fine/coarse levels, base seed b , and the deterministic step-level seed schedule (Section 3); publication packages should also retain the EC trace and selected-plan hash.

References

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